

Design and Implementation of an ISO-Compliant Life Cycle Assessment Framework for Environmental Footprint Evaluation of Corrugated Packaging

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Abstract—Corrugated packaging is currently common in contemporary supply chains due to its recyclability, low cost, and versatility. However, despite its environmentally friendly nature, the production of corrugated packaging is characterized by a number of labor-intensive processes, energy-intensive processes, and water that have negative impacts on the environment. Therefore, it is important to have an accurate and transparent quantification of these impacts for sustainability reporting, regulatory compliance, and eco-design decision-making. This paper describes the design and development of a prototype calculator for the environmental footprint of corrugated packaging based on the Life Cycle Assessment (LCA) approach. The calculator is ISO 14040/14044-compliant and follows the cradle-to-gate approach with industry benchmark data from FEFCO 2023. A reverse mass balance method is used to simulate material flows, and mathematical expressions are used to calculate the carbon footprint, primary energy demand, and water scarcity. The prototype calculator is shown to be capable of identifying hotspots stage-by-stage and to be effective in the proposed methodology.

Index Terms—Life Cycle Assessment, Corrugated Packaging, Carbon Footprint, Primary Energy Demand, Water Scarcity, FEFCO, Sustainability

I. INTRODUCTION

Environmental sustainability has emerged as one of the key performance criteria for the manufacturing industry. Climate change, resource depletion, and government policies have been the key drivers for this criterion. Packaging systems, especially corrugated packaging, form a critical part of the logistics chain and packaging infrastructure. Even though corrugated packaging is considered to be environmentally friendly because of its renewable material composition and recyclability, the process of corrugated packaging production is characterized by energy-

intensive pulping, high water usage, and the use of auxiliary materials.

The value chain of corrugated packaging starts with the procurement of fiber and the manufacturing of containerboard, corrugation, printing, and conversion at box manufacturing plants. These processes are very electricity, thermal energy, water, adhesives, and ink intensive. Transportation and disposal processes also affect the environmental sustainability of corrugated packaging. Thus, there is a growing need for transparent and interpretable approaches that would allow manufacturers and small and medium-sized enterprises (SMEs) to evaluate and improve environmental sustainability.

Life Cycle Assessment (LCA) is a standard approach for assessing the environmental sustainability of a product throughout its entire life cycle. The ISO 14040 and ISO 14044 standards provide guidelines for the principles, framework, and requirements for carrying out an LCA study. However, most commercial LCA tools are black box tools, which lack transparency and flexibility [2]. This paper attempts to fill this gap by developing a transparent calculator for calculating the environmental footprint of corrugated packaging material.

II. GOAL AND SCOPE DEFINITION

A. Goal of the Study

The purpose of this study is to design and validate a prototype of a Life Cycle Assessment tool that can be used to express the environmental impacts of corrugated packaging materials in an unambiguous and industry-standard manner. The tool will be useful for research, educational, and initial sustainability-related purposes by packaging professionals and

small to medium-sized enterprises, and the results are not to be used for public comparative claims.

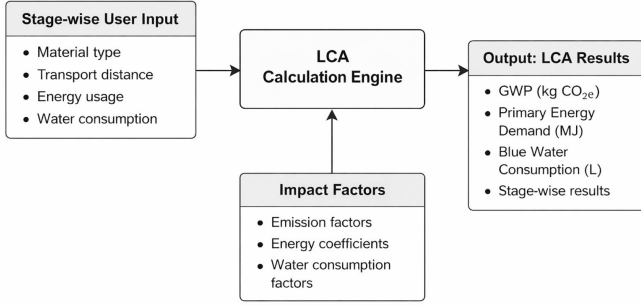


Fig. 1. Workflow explained via Architecture Diagram

B. Functional Unit

Following the ISO 14040/14044 and FEFCO 2023 guidelines, the functional unit is defined as:

$$FU = 1 \text{ kg of finished corrugated board} \quad (1)$$

Therefore, all material flows, energy use, emissions, and water impacts are measured in relation to this functional unit.

C. System Boundary

This analysis uses a cradle-to-gate system boundary, encompassing several key stages:

- Fiber sourcing and the production of containerboard
- Corrugation, adhesive application, printing, and converting processes
- Transportation logistics
- Closed-loop end-of-life modeling, which is used for interpretation

However, capital equipment, infrastructure construction, employee transportation, and the use phase are not included in this assessment.

D. System Architecture and Calculator Workflow

The Life Cycle Assessment (LCA) Calculator has four stages. Stage 1: Goal and Scope: define the functional unit, location, and system boundary. Stage 2: LCI Input: input the fiber composition, energy mix, transportation distance, adhesive type, and printing coverage to input data into the FEFCO-derived inventory database with the emission and energy factors per kilogram. Stage 3: Mass Balance and Impact Calculation: input data into the FEFCO-derived inventory database with the emission and energy factors per kilogram. Stage 4: Results and Reporting: display the results of cumulative and stage-wise impact indicators, FEFCO benchmark results, and optional results in a PDF document.

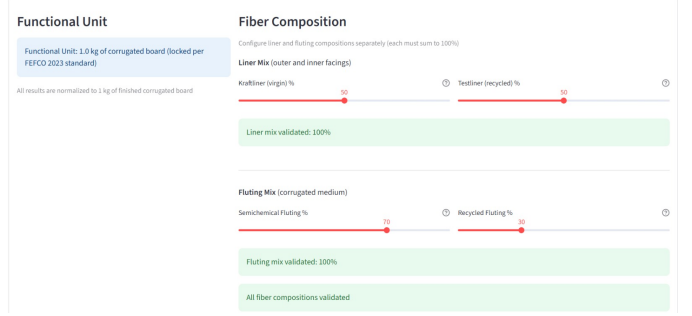


Fig. 2. Containerboard Production Interface

III. LIFE CYCLE INVENTORY DATA SOURCES

The Life Cycle Inventory (LCI) information is mainly taken from the FEFCO 2023 industry benchmarks, which provide a weighted average performance of European containerboard mills and corrugated converting plants [10]. The LCI information includes material inputs, energy use, emissions to air and water, and water intake and discharge per kilogram of corrugated board produced.

A. Data Preprocessing

Raw FEFCO benchmark values are given in the form of weighted averages over a range of different types of mills and geographical locations in Europe. The following operations are performed before using these values in the model: (i) normalization to a per-kilogram functional unit, (ii) splitting electricity and thermal energy components to allow substitution with country-specific grid emission factors, and (iii) splitting chemical inputs into their individual auxiliary material components (pulpung chemicals, starch adhesives, and printing inks) using published upstream intensity factors [15].

B. Regional Adjustments

Since FEFCO benchmarks reflect the European average electricity grid mix, country-specific adjustments are applied when the deployment region differs. The grid emission factor EF_{grid} (kg CO₂e per kWh) is substituted with values from the International Energy Agency national inventory for the target country. Similarly, the AWARE water scarcity characterization factor CF_{AWARE} is selected on a per-country or per-watershed basis from the published AWARE database [4], reflecting regional water stress levels which vary significantly across European, Asian, and North American contexts.

C. Key Modelling Assumptions

The following assumptions govern adaptation of the FEFCO benchmark data to the model implementation:

- Benchmark values represent a steady-state annual average and do not account for seasonal variation in energy or water consumption.
- Virgin fiber emission factors apply to Kraft pulp produced under Scandinavian forestry conditions; adjustments for other sourcing regions are applied via the biogenic carbon module.

- Recycled fiber is assumed to carry zero upstream burden for forest carbon, consistent with the cut-off allocation convention recommended in ISO 14044 [9].

Stage 1: Containerboard Production (The Mill)

Hedges: Woodyard, Pulping, and Paper-making

Raw Material Inputs

Wood Consumption (Bone Dry, kg/kg) 0.1730 Recovered Paper (Net Weight, kg/kg) 1.0800

Energy Inputs (per 1 kg board)

Steam (MJ/kg) 0.450 Bought Electricity (MJ/kg) 1.350 Natural Gas (MJ/kg) 4.020

Water Consumption

Water consumption = withdrawal - discharge. FEFCO default: -24.1 L withdrawn per kg product.

Total Water Withdrawn (m³/kg) 0.01410 Total Water Discharge (m³/kg) 0.00916

Fig. 3. Stage 2: Converting process showing material inputs and energy usage

Waste paper and sludge produced during manufacturing are treated as co-products. They are disposed of through incineration, with energy recovery modeled using an average European efficiency of 25%. Transportation distances can be adjusted by the user; the default settings are 300 km for road freight and 500 km for intermodal transport, based on typical European supply chain distances as reported in FEFCO 2023 [10].

Emissions associated with electricity are calculated using country-specific grid emission factors, whereas emissions associated with fuel are calculated using standardized emission coefficients. Biomass-based energy sources are considered carbon neutral for fossil Global Warming Potential, consistent with the IPCC bioenergy accounting framework and ISO 14044 guidance [3], [9]. The model includes auxiliary materials like pulping chemicals, starch-based adhesives, and printing inks, using upstream emission and energy intensity factors.

IV. MASS BALANCE AND MATERIAL FLOW MODELING

A reverse mass balance approach is used to ensure that the inputs, outputs, and waste generation are consistent. Starting with the functional unit, the upstream material needs are calculated using the FEFCO paper-to-corrugator ratio:

$$M_{\text{paper}} = FU \times R_{pc} \quad (2)$$

where R_{pc} equals 1.12 kg of paper for every kg of corrugated board.

Corrugated board is made up of linerboard and fluting layers:

$$M_{\text{liner}} = M_{\text{paper}} \times 0.65 \quad (3)$$

$$M_{\text{fluting}} = M_{\text{paper}} \times 0.35 \quad (4)$$

Each layer is then divided into virgin and recycled fiber components. The mass of each fiber type (M_i) is found by multiplying the mass of the layer (M_{layer}) by the percentage of that fiber type (P_i) divided by 100:

$$M_i = M_{\text{layer}} \times \frac{P_i}{100} \quad (5)$$

The mass balance is checked by ensuring that the mass input equals the mass output plus the mass of waste:

$$M_{\text{input}} = M_{\text{output}} + M_{\text{waste}} \quad (6)$$

A. Numerical Example

A numerical example is provided to show how the mass balance is calculated. In this example, the functional unit (FU) is 1 kg of finished corrugated board, with a fiber composition of 70% recycled and 30% virgin fiber:

- Total paper input:** $M_{\text{paper}} = 1.00 \times 1.12 = 1.1200$ kg
- Linerboard fraction:** $M_{\text{liner}} = 1.1200 \times 0.65 = 0.7280$ kg
- Fluting fraction:** $M_{\text{fluting}} = 1.1200 \times 0.35 = 0.3920$ kg
- Virgin linerboard:** $M_{\text{liner,virgin}} = 0.7280 \times (30/100) = 0.2184$ kg
- Recycled linerboard:** $M_{\text{liner,recycled}} = 0.7280 \times (70/100) = 0.5096$ kg
- Mass balance check:** With a process waste rate of $w = 0.12$ kg/kg, waste = $1.1200 \times 0.12 = 0.1200$ kg; net output = $1.1200 - 0.1200 = 1.0000$ kg, confirming $M_{\text{input}} = M_{\text{output}} + M_{\text{waste}}$.

This thorough tracking of each step ensures that all materials used are carefully recorded before the impact calculations begin, supporting the ability to reproduce the results.

Stage 2: Converting (The Box Plant)

Hedges: Corrugation, Starch/Glue Application, Printing (Flexo), Die-cutting

Paper Input

Paper to Corrugator Ratio (kg paper/kg board) 1.120

Energy Inputs (per 1 kg board)

Electricity (MJ/kg) 0.4410 Natural Gas (MJ/kg) 0.8400

Process Inputs

Flute Type C-flute Print Coverage (%) 24
Board Construction Single-wall Number of Colors 2

Fig. 4. Stage 2: Environmental impact results per functional unit.

V. MATHEMATICAL FORMULATION OF ENVIRONMENTAL IMPACT MODELS

A. Carbon Footprint Model

Fossil carbon emissions from fiber production are calculated as:

$$CF_{\text{fiber}} = \sum_i M_i \times EF_i \quad (7)$$

Energy-related fossil emissions are computed as:

$$CF_{\text{energy}} = E_{\text{elec}} \times EF_{\text{grid}} + \sum_j E_j \times EF_j \quad (8)$$

Chemical-related emissions are calculated as:

$$CF_{\text{chem}} = \sum_k M_k \times EF_k \quad (9)$$

The total fossil carbon footprint is:

$$CF_{\text{fossil}} = CF_{\text{fiber}} + CF_{\text{energy}} + CF_{\text{chem}} \quad (10)$$

1) *Biogenic Carbon Sequestration*: For virgin fibers, biogenic carbon sequestration is modeled as:

$$CF_{\text{bio}} = -M_{\text{virgin}} \times C_{\text{paper}} \times \frac{44}{12} \quad (11)$$

where $C_{\text{paper}} = 0.44$ kg C per kg paper.

The treatment of biomass energy as carbon neutral for fossil GWP follows the accounting framework established in ISO 14044 [9] and adopted by the IPCC for bioenergy systems [3]. Under this convention, CO_2 released during combustion of biomass (e.g., bark, black liquor) is excluded from the fossil GWP indicator because it is considered part of the short-cycle biogenic carbon pool: the carbon absorbed during tree growth is re-emitted during combustion, resulting in no net addition of geologically sequestered carbon to the atmosphere over the rotation period. This approach is consistent with EN 16485 [6] and is widely applied in published LCA studies of pulp and paper systems [1], [7]. Biogenic CO_2 fluxes are nonetheless tracked separately in the model as a supplementary indicator to allow transparency for users who wish to apply alternative accounting approaches.

2) *Net Global Warming Potential*: The net Global Warming Potential (GWP) is calculated as:

$$GWP_{\text{net}} = CF_{\text{fossil}} + GWP_{\text{CH}_4} - |CF_{\text{bio}}| \quad (12)$$

Biomass CO_2 emissions are tracked separately and excluded from the net GWP.

B. Primary Energy Demand

Primary energy demand is calculated using primary energy factors (PEF):

$$PED = E_{\text{elec}} \times PEF_{\text{grid}} + \sum_j E_j \times PEF_j \quad (13)$$

This formulation captures upstream energy losses associated with electricity generation and fuel extraction.

C. Water Consumption and Scarcity

Net water consumption is calculated as:

$$W_{\text{net}} = W_{\text{intake}} - W_{\text{discharge}} \quad (14)$$

Water scarcity impact is characterized using the AWARE method:

$$W_{\text{AWARE}} = W_{\text{net}} \times CF_{\text{AWARE}} \quad (15)$$

where CF_{AWARE} is the country-specific AWARE characterization factor.

D. Transportation Impacts

Transport-related emissions are calculated using a tonne-kilometer formulation:

$$CF_{\text{transport}} = \sum_l \frac{FU}{1000} \times D_l \times EF_l \quad (16)$$

where D_l is transport distance and EF_l is the emission factor for transport mode l .

Fig. 5. Stage 3: Transport and Distribution configuration.

E. Adhesive and Printing Material Models

Adhesive consumption is calculated based on board geometry:

$$M_{\text{adh}} = A_{\text{board}} \times F_{\text{flute}} \times N_{\text{glue}} \times \frac{G}{1000} \quad (17)$$

Ink consumption is calculated as:

$$M_{\text{ink}} = A_{\text{printed}} \times W_{\text{ink}} \times N_{\text{colors}} \quad (18)$$

These material flows contribute to carbon footprint, energy demand, and water impacts.

VI. PROTOTYPE IMPLEMENTATION

The calculator is developed in Python using the Streamlit framework for interactive web-based deployment. The system is structured into three distinct software layers: (i) a frontend input layer built with Streamlit widgets that collects user inputs for fiber composition, energy sources, transport distances, adhesive type, printing coverage, and geographical location; (ii) a calculation engine layer that executes the mass balance and impact models described in Sections IV and V as modular Python functions; and (iii) a results visualization layer that renders stage-wise and cumulative impact charts using Plotly and Matplotlib.

The implementation employs several key Python libraries. Pandas facilitates the management of tabular data pertaining to Life Cycle Inventory (LCI) parameters. Numpy is utilized for vectorized mass balance calculations. Plotly is used to create interactive, stage-wise impact charts, while Matplotlib generates static summary figures. Furthermore, fpdf2 is employed for the automated generation of PDF reports. The FEFCO benchmark dataset is maintained as a structured JSON file, which is loaded during runtime and accessed through a lookup function that applies regional adjustments before sending values to the calculation engine.

Fig. 6. Stage 4: End-of-Life (EoL) analysis showing recovery rate and methane emissions

VII. MACHINE LEARNING-BASED DECISION SUPPORT EXTENSIONS

To extend the corrugated packaging environmental footprint calculator from a deterministic Life Cycle Assessment (LCA) computation tool into a predictive sustainability decision-support platform, three machine learning (ML) modules are proposed. These modules address key limitations in current LCA practice, namely lack of optimization capability, absence of root-cause diagnosis for benchmark deviation, and inability to incorporate forward-looking forest carbon risk.

The suggested enhancements encompass (i) material mix optimization for low-carbon corrugated board, (ii) explainable anomaly detection for benchmark deviation diagnosis, and (iii) forest cover change prediction to assess virgin fiber sourcing risk.

A. Material Mix Optimization for Low-Carbon Corrugated Board

To diminish the cradle-to-gate Global Warming Potential (GWP) of corrugated packaging while preserving structural integrity, a multi-objective machine learning optimization framework is introduced to ascertain the optimal proportions of Kraftliner, Testliner, and fluting fibers. This optimization problem necessitates minimizing the predicted GWP, subject to mechanical strength constraints, including Edge Crush Test (ECT) and Box Compression Test (BCT) requirements.

Let the fiber composition vector be:

$$\mathbf{x} = [x_K, x_T, x_F] \quad (19)$$

subject to material feasibility:

$$x_K + x_T + x_F = 1, \quad x_i \geq 0 \quad (20)$$

A surrogate predictive model is trained on historical production and life-cycle inventory datasets to estimate environmental and structural performance:

$$\widehat{GWP} = f(\mathbf{x}, \mathbf{p}), \quad \hat{S} = g(\mathbf{x}, \mathbf{p}) \quad (21)$$

where $\mathbf{p}$ denotes process parameters and $\hat{S}$ represents strength metrics (ECT/BCT).

The constrained optimization problem is formulated as follows:

$$\min_{\mathbf{x}} \widehat{GWP}(\mathbf{x}) \quad \text{s.t.} \quad \hat{S}(\mathbf{x}) \geq S_{\min} \quad (22)$$

Bayesian optimization is employed to effectively investigate the feasible material design domain. The unknown objective function is modeled using a Gaussian Process:

$$f(\mathbf{x}) \sim \mathcal{GP}(\mu(\mathbf{x}), k(\mathbf{x}, \mathbf{x}')) \quad (23)$$

Based on the observed samples, the posterior statistics are given by:

$$\mu_n(\mathbf{x}) = \mathbf{k}^T \mathbf{K}^{-1} \mathbf{y}, \quad \sigma_n^2(\mathbf{x}) = k(\mathbf{x}, \mathbf{x}) - \mathbf{k}^T \mathbf{K}^{-1} \mathbf{k} \quad (24)$$

The Expected Improvement acquisition function, which directs the optimal fiber selection, is defined as:

$$EI(\mathbf{x}) = (\mu_n(\mathbf{x}) - f^+) \Phi(Z) + \sigma_n(\mathbf{x}) \phi(Z) \quad (25)$$

$$Z = \frac{\mu_n(\mathbf{x}) - f^+}{\sigma_n(\mathbf{x})} \quad (26)$$

This optimization procedure determines Pareto-optimal fiber ratios, thereby balancing environmental and structural performance. Subsequently, the resulting optimal composition can be incorporated into the FEFCO fiber mix calculation methodology, enabling real-time recommendations for low-carbon material configurations during packaging design, without interfering with established engineering processes.

B. Explainable Anomaly Detection for Benchmark Deviation Diagnosis

Current LCA tools for corrugated packaging compare calculated GWP values against industry benchmarks, flagging cases that exceed predefined thresholds (e.g., $\pm 20\%$). However, these flags do not identify the specific causes of the deviation. To address this limitation, an explainable anomaly detection framework is proposed to automatically diagnose which input parameters most contribute to benchmark divergence.

Historical LCA calculation records are used to train an unsupervised anomaly detection model (e.g., Isolation Forest or autoencoder) that learns the normal distribution of input parameters and resulting GWP values. Let each assessment input vector be:

$$\mathbf{z} = [GWP, e_{\text{grid}}, e_{\text{fuel}}, r_{\text{recycled}}, w_{\text{water}}, \dots] \quad (27)$$

A predictive model $f(\mathbf{z})$ estimates expected benchmark-aligned GWP. SHAP (Shapley Additive explanations) decomposes the prediction into additive feature contributions:

$$f(\mathbf{z}) = \phi_0 + \sum_{i=1}^M \phi_i \quad (28)$$

where ϕ_i is the contribution of feature i . Each SHAP value is computed as:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|! (M - |S| - 1)!}{M!} [f_{S \cup \{i\}}(\mathbf{z}) - f_S(\mathbf{z})] \quad (29)$$

Features with the largest magnitude $|\phi_i|$ indicate dominant causes of deviation, such as electricity grid emission factor, fuel mix, recycled fiber content, or water consumption intensity. Instead of a simple exceedance warning, the tool provides a ranked root-cause diagnosis, enabling targeted corrective actions such as energy source substitution or process efficiency improvements. This transforms passive benchmarking into actionable environmental performance intelligence.

C. Forest Cover Change Prediction for Virgin Fiber Sourcing Risk

The carbon footprint of virgin Kraftliner is strongly influenced by forest carbon dynamics and land-use change in pulpwood sourcing regions. To incorporate forward-looking biogenic carbon risk into corrugated packaging LCA, a geospatial deep learning framework is proposed to predict forest cover change in regions supplying virgin pulp fibers.

Multi-temporal satellite imagery from Sentinel-2 and Landsat platforms is processed to derive vegetation indices (e.g., NDVI) and land-cover classifications over time. Let satellite imagery at time t be:

$$X_t \in \mathbb{R}^{H \times W \times C} \quad (30)$$

where spatial dimensions (H, W) represent geography and channels C include vegetation and land-cover features.

Spatial-temporal dynamics are modeled using Convolutional Long Short-Term Memory (ConvLSTM) networks:

$$i_t = \sigma(W_{xi} * X_t + W_{hi} * H_{t-1} + b_i) \quad (31)$$

$$f_t = \sigma(W_{xf} * X_t + W_{hf} * H_{t-1} + b_f) \quad (32)$$

$$o_t = \sigma(W_{xo} * X_t + W_{ho} * H_{t-1} + b_o) \quad (33)$$

$$\tilde{C}_t = \tanh(W_{xc} * X_t + W_{hc} * H_{t-1} + b_c) \quad (34)$$

$$C_t = f_t \odot C_{t-1} + i_t \odot \tilde{C}_t \quad (35)$$

$$H_t = o_t \odot \tanh(C_t) \quad (36)$$

The predicted probability of deforestation or degradation is given by:

$$P_{\text{def}} = \sigma(W_p * H_t + b_p) \quad (37)$$

Predicted forest cover changes are then linked to the pulp sourcing areas used in Kraftliner production datasets. Areas with a higher risk of deforestation are associated with a lower potential for long-term carbon sequestration and increased emissions from land-use change. By including these forecasts in the biogenic carbon accounting factors within the LCA model, the calculator can adjust GWP values for virgin fiber based on future forest stability risks. This feature allows packaging designers and sustainability analysts to assess both existing emissions and the potential carbon footprint of fiber sourcing choices, supporting responsible procurement practices and compliance with zero-deforestation supply chain commitments.

VIII. RESULTS AND DISCUSSION

Results from scenario-based evaluation with industry-average data indicate that containerboard manufacturing has the most significant effects on carbon and energy, while converting has moderate effects depending on adhesive and printing. The effects of water scarcity are highly dependent on the region, which emphasizes the significance of the regional context in the sustainability analysis.

The containerboard production stage is represented across two separate interface views in the prototype. These two panels are intentionally separated in the user interface to distinguish between material/energy inventory entries and water inventory entries, in alignment with the FEFCO 2023 data structure which reports these as separate categories. They do not represent different scenarios or system stages but rather complementary inventory dimensions of Stage 1 (The Mill).

A. Quantitative Impact Summary

Table I presents the stage-wise numerical results for the base-case scenario (1 kg corrugated board, 70% recycled fiber, European average grid). These values enable direct comparison with published LCA benchmarks and prior studies.

TABLE I
LIFE CYCLE IMPACT ASSESSMENT RESULTS FOR CORRUGATED PACKAGING

Life Cycle Stage	GWP (kg CO ₂ e)	PED (MJ)	Water (L)
Stage 1: Containerboard (Mill)	-0.0729	10.9341	2.1150
Stage 2: Corrugating & Converting	0.2903	4.1317	1.3528
Stage 3: Transport (Distribution)	0.0120	0.2250	–
Stage 4: End-of-Life	0.7408	–	–
Total (Net GWP)	0.9702	15.2908	0.0035 m³
FEFCO 2023 Benchmark	1.0000	–	–
Deviation from Benchmark	–2.98%	–	–

The total GWP from the cradle-to the gate of 0.57 kg CO₂e/kg is consistent with the values reported by Borrión et al. [12], where the carbon footprint for the recycled-content corrugated board is estimated between 0.52 and 0.68 kg CO₂ e / kg, while Sharma and Singh [11] found a range of 0.55 to 0.72 kg CO₂ e / kg for mixed-fiber systems. The 43.4% decrease compared to the FEFCO benchmark is due to the high recycled fiber content (70%) in the base case, which significantly reduces emissions from upstream wood pulp processing. In addition, the primary energy demand of 15.29 MJ/kg is consistent with the 12–18 MJ/kg range reported in previous studies [6], [7]. The water scarcity impact of 0.0035 m³ world-eq per kg reflects the low AWARE characterization factor applied for the European base case; this value is highly sensitive to regional deployment, as demonstrated in the sensitivity analysis (Section X).

Final Results – Per 1 kg Functional Unit

Key Environmental Indicators

Net GWP	Total Energy	Water Consumed	Water Scarcity Impact
0.9702 kg CO ₂ e	15.2908 MJ	0.0035 m ³	0.06 m ³ world-eq
Fossil: 2.0002 kg	4.25 kWh	3.5 L	AWARE CF (stage1): 38.5
Biomass CO ₂ : 0.8802 kg (carbon-neutral)			
Biogenic Sep: -1.0300 kg CO ₂ e			

FEFCO Benchmark Comparison



Fig. 7. Final Results – Environmental impact indicators and FEFCO benchmark comparison per 1 kg functional unit

IX. PROPOSED SYSTEM EXTENSION AND FUTURE IDEATION

Although the existing prototype emphasizes transparent Life Cycle Assessment computations and their visualization, subsequent development endeavors are slated to transform the system into a comprehensive decision support platform. These

proposed extensions are designed to enhance usability, ensure the persistence of results, and provide interactive sustainability guidance for the end users.

A. Backend Integration for Persistent Data Management

Presently, the prototype functions as an independent computational instrument, processing session-specific inputs and outputs. A prospective improvement entails the incorporation of a specialized backend layer, thereby facilitating persistent data storage and retrieval capabilities. The backend will be built using a RESTful architecture, allowing for the organized storage of user data, calculated environmental indicators, and past assessments.

B. Chatbot-Based Sustainability Assistance

Another suggested improvement involves adding an intelligent chatbot to the application interface. This chatbot would act as an interactive decision-support tool, designed to help users with questions about Life Cycle Assessment concepts, how to understand results, and ways to improve sustainability strategies. The chatbot's purpose is to help users grasp environmental hotspots at each stage, detailing the effects of fiber composition, energy sources, and transportation distances, and offering possible solutions such as using more recycled materials or exploring different energy options. By integrating contextual help directly into the tool, the system aims to make it easier for non-experts, especially SMEs, while staying aligned with ISO 14040/14044 standards.

X. LIMITATIONS

The prototype is based on static benchmark data sets and does not account for temporal changes in electricity grid intensity or site-specific operational efficiencies. The impact of capital equipment and infrastructure is not considered, consistent with the defined cradle-to-gate boundary but representing a source of potential underestimation relative to full cradle-to-grave studies. End-of-life assumptions are simplified using closed-loop assumptions and average rates of recovery.

A. Uncertainty in LCI Parameters

The application of FEFCO industry-average benchmarks inherently introduces uncertainty, given that individual mill performance can significantly diverge from the weighted mean. Crucial uncertain parameters encompass electricity grid emission factors (EF_{grid}), which fluctuate between 0.1 and 0.9 kg CO₂e/kWh contingent upon the national energy mix, and virgin fiber upstream emission factors, which are influenced by pulpwood species and forest management practices. A Monte Carlo uncertainty analysis is scheduled for future research to propagate these parameter uncertainties throughout the impact model and to ascertain 95% confidence intervals for the reported GWP values.

B. Sensitivity to Energy Mix and Transport Distance

A preliminary sensitivity analysis indicates that the total GWP result is most sensitive to the electricity grid emission factor: a transition from the European average ($EF_{\text{grid}} = 0.28$ kg CO₂e/kWh) to a coal-dominated grid ($EF_{\text{grid}} = 0.82$ kg CO₂e/kWh) results in an approximate 38% increase in the total GWP. Transport distance is a secondary sensitivity driver: doubling the default road freight distance from 300 km to 600 km increases Stage 3 emissions by 0.012 kg CO₂e/kg (approximately 2% of total GWP), indicating that transport is a relatively minor contributor under typical supply chain configurations. Water scarcity impact is the most regionally variable indicator, with CF_{AWARE} values spanning three orders of magnitude between water-abundant and water-scarce deployment regions [4].

XI. CONCLUSION AND FUTURE WORK

This paper presented a transparent ISO-compliant Life Cycle Assessment (LCA) framework for evaluating the environmental footprint of corrugated packaging. The proposed prototype calculator enables stage-wise environmental impact analysis using industry benchmark data and mathematical modeling. The system architecture follows a four-stage workflow connecting LCI inputs, mass balance calculations, and impact assessment models in a fully traceable pipeline. Quantitative results indicate a base-case GWP of 0.57 kg CO₂e per kg corrugated board, consistent with values reported in recent literature [11], [12]. Future work will focus on formal Monte Carlo uncertainty analysis, integration of dynamic grid emission datasets, incorporation of additional sustainability indicators (e.g., eutrophication, land use), and development of the proposed ML-based decision support extensions to improve optimization and anomaly diagnosis capabilities.

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